

Melissa Aoko

1. Topic Introduction

Economic growth and development have remained central priorities for governments and international organizations worldwide. Gross Domestic Product (GDP) serves as the most widely recognized measure of a country's economic output, reflecting the total monetary value of all goods and services produced within a given period. Understanding GDP trends, growth rates, and per capita income distributions is essential for evaluating national development, identifying economic disparities, and informing policy decisions.

This project focuses on analyzing GDP growth trends, GDP per capita disparities, and income levels across 262 countries and territories from 2000 to 2024. The analysis was carried out using Power BI to create an interactive dashboard that communicates global economic patterns through data visualization and storytelling.

Countries covered span all major world regions including North America, Europe, Asia, Africa, South America, and Oceania, providing a comprehensive global view of economic performance across both developed and developing economies.

Data Source

The dataset was obtained from the **World Bank Open Data platform** and included the following indicators:

- GDP (current US\$)
- GDP growth rate (annual %)
- GDP per capita (current US\$)
- GNI per capita (current US\$)

The use of multiple indicators allowed for a broader understanding of how economic output, income distribution, and growth rates vary across countries and over time.

2. Analytical Objectives and Questions

The project was guided by the following analytical objectives and questions:

Objective 1: Analyze GDP growth patterns across countries over time. **Question:** How has global average GDP growth changed between 2000 and 2024?

Objective 2: Examine GDP per capita disparities between nations. **Question:** Which countries have the highest GDP per capita and what does this reveal about global income inequality?

Objective 3: Investigate the relationship between GNI per capita and GDP growth rate.

Question: Does a higher GNI per capita correspond with a higher or lower GDP growth rate?

Objective 4: Compare the share of global GDP across countries. **Question:** Which countries dominate global GDP output?

Objective 5: Analyze how global economic shocks have affected GDP growth trends.

Question: How did events such as the 2008 financial crisis and COVID-19 pandemic affect GDP growth across countries?

The selected variables aligned well with these objectives because they represent the most widely used economic indicators in national income and development analysis.

3. Methods Used

Data Collection

The datasets were downloaded from the World Bank Open Data platform. The data covered 262 countries and territories with yearly observations from 2000 to 2024.

Data Cleaning and Preparation

Power Query within Power BI was used to clean and prepare the data. The cleaning process involved:

- Removing unnecessary metadata rows and columns added by the World Bank export format
- Renaming variables for clarity (e.g. Attribute → Year, Value → indicator name)
- Converting data types appropriately (Year as Whole Number, values as Decimal Number)
- Restructuring the datasets from wide format into long format using the Unpivot Columns function
- Filtering out null and blank values
- Merging all four indicator tables using Country Name and Year into a single master table

Exploratory Data Analysis (EDA)

Exploratory analysis was performed to identify trends, patterns, and anomalies. Particular attention was paid to temporal changes in GDP growth, country-level income comparisons, and the relationship between GNI per capita and growth rate.

Dashboard Development

An interactive Power BI dashboard was developed containing three KPI cards, a line chart, a bar chart, a map visual, a scatter plot, and a treemap. Interactivity was enabled through a Year range slicer (2000–2024) and a Country Name dropdown slicer.

4. Key Insights and Findings

Global Average GDP Growth

The dashboard KPI card shows a global average GDP growth rate of **3.44%** across all 262 countries over the study period. The line chart tracking GDP growth over time reveals significant volatility, with notable downturns visible around **2008–2009** corresponding to the global financial crisis, and again around **2020** during the COVID-19 pandemic. These two events represent the most significant disruptions to global economic growth in the modern era.

GDP Per Capita Disparities

The bar chart highlights extreme income inequality across nations. The highest GDP per capita countries are **Liechtenstein, Luxembourg, Bermuda, Norway, Switzerland, Qatar, and Ireland**, all recording values significantly above the global average of **\$14,910 USD**. These countries benefit from strong financial sectors, small populations, and high-value industries. This stands in sharp contrast to lower-income nations across Africa and South Asia, illustrating the wide development gap that persists globally.

GNI Per Capita vs GDP Growth Rate

The scatter plot comparing GNI per capita against GDP growth rate reveals that countries with very high GNI per capita tend to cluster at lower and more stable growth rates. This pattern is consistent with economic convergence theory, which suggests that wealthier economies grow more slowly as they approach their productive limits, while lower-income economies have greater potential for rapid growth. The wide spread of data points also suggests that factors beyond income level such as governance, trade, and natural resources do play important roles in determining growth outcomes.

Global GDP Share

The treemap confirms that global GDP output is heavily concentrated among a small number of economies. The **United States and China** dominate as the two largest blocks, followed by **Japan, Germany, the United Kingdom, France, and India**. Together these countries account for the majority of global economic output, while the remaining 250+ countries share a comparatively small portion. This concentration reflects historical advantages in industrialization, technology, and institutional development.

5. Recommendations

Based on the findings of this dashboard analysis, the following recommendations are proposed:

Countries with persistently high GDP per capita such as Luxembourg and Norway should serve as benchmarks for sustainable economic policy, particularly regarding investment in human capital, institutional quality, and diversified industries. Developing economies should study and adapt these models to their own contexts.

The significant GDP per capita gap between high-income and low-income nations calls for continued international development support and fairer global trade policies that give developing countries greater access to markets and financial resources.

Finally, policymakers should look beyond GDP growth as a single measure of success and incorporate indicators such as GNI per capita and income equality to ensure that economic growth translates into improved living standards for all citizens.

6. Conclusion

This project successfully demonstrated how global economic growth indicators can be explored and communicated using interactive dashboard techniques in Power BI. The dashboard transformed raw World Bank data covering 262 countries from 2000 to 2024 into clear and meaningful visual insights.

The analysis revealed a global average GDP growth rate of 3.44%, significant income disparities between nations and the dominance of the United States and China in global GDP share. Countries with the highest GDP per capita were concentrated in Europe and the Middle East, while global growth volatility was most pronounced during crisis periods.

The use of interactive slicers, KPI cards, maps, and multiple chart types allowed for dynamic exploration of the data, demonstrating the power of visual analytics in communicating complex economic findings in an accessible and engaging way.

Overall, this project highlights the important role of data analytics in understanding global economic development and supporting evidence-based decision-making in policy, investment, and international development.